MATH 128A - HW 1

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Problem 1: Preliminaries

(a) Yes, there will be a final exam for this class. It is scheduled for 12/16/2025 from 3-6pm.

(b) The lowest score of the three programming assignments will be dropped.

(c) Yes, there will be biweekly quizzes in this class. They will take place in weeks 3, 5, 7, 9, 11, and 15.

(d) I can find announcements and materials related to the course on bCourses (Canvas), which is the official course website.

(e) I should ask math-related questions on Ed Discussion. For more direct communication, I can also bring them up during lectures or in office hours.

Problem 2: Problems from textbook Section 1.1

1b

By the Intermediate Value Theorem: define the function $f(x) = (x-2)^2 - \ln x$. Since $(x-2)^2$ is a polynomial and $\ln x$ is continuous for $x > 0$, $f(x)$ is continuous for $x > 0$. Both intervals $[1, 2]$ and $[e, 4]$ are within $x > 0$, so f is continuous on these intervals.

For the interval $[1, 2]$:

- At $x = 1$: $f(1) = (1-2)^2 - \ln 1 = 1 - 0 = 1 > 0$.
- At $x = 2$: $f(2) = (2-2)^2 - \ln 2 = -\ln 2 \approx -0.693 < 0$

Since $f(1) > 0$ and $f(2) < 0$, f changes sign on $[1, 2]$. By the Intermediate Value Theorem, there exists at least one $c \in (1, 2)$ such that $f(c) = 0$.

For the interval $[e, 4]$:

- At $x = e$: $f(e) = (e-2)^2 - \ln e = (e-2)^2 - 1 \approx -0.485 < 0$
- At $x = 4$: $f(4) = (4-2)^2 - \ln 4 = 4 - \ln 4 \approx 2.614 > 0$

Since $f(e) < 0$ and $f(4) > 0$, f changes sign on $[e, 4]$. By the Intermediate Value Theorem, there exists at least one $d \in (e, 4)$ such that $f(d) = 0$.

Thus, the equation $(x-2)^2 - \ln x = 0$ has at least one solution in each of the intervals $[1, 2]$ and $[e, 4]$.

5b

Check the Endpoints

- At $x = 0.5$: $f(0.5) = \frac{4(0.5)-3}{(0.5)^2-2(0.5)} = \frac{4}{3}$
- At $x = 1$: $f(1) = \frac{4(1)-3}{1^2-2(1)} = \frac{1}{-1} = -1$

Check for Critical Points

Since $f(x)$ is a rational function and the denominator $x(x-2) \neq 0$ on $[0.5, 1]$, the function is continuous on this interval.

Compute the derivative using the quotient rule:

$$f'(x) = \frac{(4)(x^2-2x) - (4x-3)(2x-2)}{(x^2-2x)^2} = \frac{-4x^2+6x-6}{(x^2-2x)^2}$$

The denominator is always positive on $[0.5, 1]$. The quadratic $-4x^2+6x-6$ has discriminant:

$$\Delta = 6^2 - 4(-4)(-6) = 36 - 96 = -60 < 0$$

So the numerator is always negative $\implies f'(x) < 0$ on $[0.5, 1] \implies f(x)$ is strictly decreasing.

- On $[0.5, 0.75]$: $f(x) > 0$ and decreasing $\implies |f(x)|$ decreases from $\frac{4}{3}$ to 0
- On $[0.75, 1]$: $f(x) < 0$ and decreasing $\implies |f(x)| = -f(x)$ increases from 0 to 1

Final Answer

The maximum value of $|f(x)|$ occurs at the left endpoint:

$$|f(0.5)| = \boxed{\frac{4}{3}}$$

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Suppose, for contradiction, that there exist two distinct numbers p and q in $[a, b]$ such that $f(p) = 0$ and $f(q) = 0$, with $p < q$.

Since f is continuous on $[a, b]$ and differentiable on (a, b), and $f(p) = f(q) = 0$, by **Rolle's Theorem**, there exists some $c \in (p, q) \subset (a, b)$ such that:

$$f'(c) = 0$$

This contradicts the given condition that $f'(x) \neq 0$ for all $x \in (a, b)$.

Therefore, the assumption that there are two distinct zeros is false. Hence, there can be **at most one** number p in $[a, b]$ such that $f(p) = 0$.

11 (abcd)

The second Taylor polynomial $P_2(x)$ for $f(x) = e^x \cos x$ about $x_0 = 0$ is found by computing the function and its first two derivatives at $x = 0$:

$$\begin{aligned} f(x) &= e^x \cos x, & f(0) &= 1 \\ f'(x) &= e^x(\cos x - \sin x), & f'(0) &= 1 \\ f''(x) &= -2e^x \sin x, & f''(0) &= 0 \end{aligned}$$

Thus,

$$P_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 = 1 + x.$$

(a) $P_2(0.5) = \boxed{1.5}$, which approximates $f(0.5)$.

The Taylor remainder is $R_2(x) = \frac{f'''(c)}{6}x^3$ for some c between 0 and x, where $f'''(x) = -2e^x(\sin x + \cos x)$. For $x = 0.5 > 0$, the error bound is $|f(0.5) - P_2(0.5)| \le \frac{M}{6}(0.5)^3$, where $M = \max_{c\in[0,0.5]} |f'''(c)| = 2e^{0.5}(\sin 0.5 + \cos 0.5)$ (as this is increasing on $[0, 0.5]$). Numerically, $M \approx 4.474$, so the bound is approximately $\boxed{0.093}$.

The actual value is $f(0.5) \approx 1.447$, so the actual error is approximately 0.053. The bound exceeds the actual error.

(b) On $[0, 1]$, $|f(x) - P_2(x)| \le \frac{M}{6}x^3 \le \frac{M}{6}$, where $M = \max_{x\in[0,1]} |f'''(x)| = 2e(\sin 1 + \cos 1)$ (as this is increasing on $[0, 1]$). Numerically, $M \approx 7.512$, so the bound is approximately $\boxed{1.252}$.

(c) $\int_0^1 P_2(x)dx = \int_0^1 (1 + x)dx = \left[x + \frac{x^2}{2}\right]_0^1 = \boxed{1.5}$, which approximates $\int_0^1 f(x)dx$.

(d) The error satisfies

$$\left|\int_0^1 f(x)dx - \int_0^1 P_2(x)dx\right| = \left|\int_0^1 R_2(x)dx\right| \le \int_0^1 |R_2(x)|dx \le \int_0^1 \frac{M}{6}x^3dx = \frac{M}{6}\cdot\frac{1}{4} = \frac{M}{24},$$

where $M \approx 7.512$ from part (b). Numerically, the bound is approximately $\boxed{0.313}$.

The antiderivative of $f(x)$ is $\frac{1}{2}e^x(\sin x + \cos x)$, so $\int_0^1 f(x)dx = \frac{1}{2}\left[e(\sin 1 + \cos 1) - 1\right] \approx 1.378$. The actual error is approximately $\boxed{0.122}$. The bound exceeds the actual error.

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$P_2(x)$ is a degree-2 polynomial. The third derivative of $\cos x$ is zero at $x = 0$. The Taylor series for $\cos x$ contains only even powers. Therefore, the next nonzero term is of degree 4.

By Taylor's Theorem, there exists ξ between 0 and x such that:

$$\cos x = P_3(x) + R_3(x), \quad \text{where} \quad R_3(x) = \frac{f^{(4)}(\xi)}{4!}x^4.$$

$$\because f(x) = \cos x \Rightarrow f^{(4)}(x) = \cos x.$$

$$\therefore |f^{(4)}(\xi)| = |\cos\xi| \le 1.$$

Since $|x| \le \frac{1}{2}$:

$$|x|^4 \le \left(\frac{1}{2}\right)^4 = \frac{1}{16}.$$

Therefore,

$$|R_3(x)| = \left|\frac{\cos\xi}{24}x^4\right| \le \frac{1}{24}\cdot\frac{1}{16} = \boxed{\frac{1}{384} \approx 0.0026}$$

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(a) Derivation of the Series

The Maclaurin series for e^{-t^2} is

$$e^{-t^2} = \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{k!}.$$

Integrating term by term from 0 to x,

$$\int_0^x e^{-t^2}dt = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}\int_0^x t^{2k}dt = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}\left[\frac{t^{2k+1}}{2k+1}\right]_0^x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)k!}.$$

Multiplying by $\frac{2}{\sqrt{\pi}}$,

$$\operatorname{erf}(x) = \boxed{\frac{2}{\sqrt{\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)k!}}$$

(c) Approximation of erf(1)

For $x = 1$, the series is

$$\operatorname{erf}(1) = \frac{2}{\sqrt{\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)k!}.$$

This is an alternating series with terms decreasing in absolute value to 0. The error after summing the first $n+1$ terms (up to $k = n$) is less than the absolute value of the next term (for $k = n+1$).

The absolute value of the term for k is $\frac{2}{\sqrt{\pi}} \cdot \frac{1}{(2k+1)k!}$. Computing terms shows:

- For $k = 9$, the term has absolute value $\approx 1.64 \times 10^{-7}$.
- For $k = 10$, the term has absolute value $\approx 1.48 \times 10^{-8} < 10^{-7}$.

Thus, sum up to $k = 9$ (10 terms total) to ensure the error is less than 10^{-7}. The partial sum is 0.84270077928364132442, which approximates erf(1) to within 10^{-7}. Rounded to 7 decimal places (consistent with the error bound), $\boxed{\operatorname{erf}(1) \approx 0.8427008}$.

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(a) Lipschitz ⇒ continuous

Assume f is Lipschitz on $[a, b]$ with constant L. For any $\varepsilon > 0$ choose $\delta = \varepsilon/L$ (if $L = 0$ take δ arbitrary). Then for any $x, y \in [a, b]$ with $|x - y| < \delta$,

$$|f(x) - f(y)| \leq L|x - y| < L \cdot \frac{\varepsilon}{L} = \varepsilon.$$

Hence f is (uniformly) continuous on $[a, b]$. In particular $f \in C[a, b]$.

(b) Bounded derivative ⇒ Lipschitz

Suppose f is differentiable on $[a, b]$ and $|f'(x)| \leq L$ for all $x \in [a, b]$. For any $x, y \in [a, b]$ the Mean Value Theorem gives a point ξ between x and y with

$$f(x) - f(y) = f'(\xi)(x - y).$$

Taking absolute values and using the bound on f',

$$|f(x) - f(y)| = |f'(\xi)||x - y| \leq L|x - y|.$$

Thus f satisfies a Lipschitz condition with constant L.

(c) Example: continuous but not Lipschitz

Take $f(x) = \sqrt{x}$ on $[0, 1]$. This function is continuous on $[0, 1]$ but not Lipschitz there. To see it is not Lipschitz, suppose for contradiction there were $M > 0$ with

$$|\sqrt{x} - \sqrt{0}| = \sqrt{x} \leq M|x - 0| = Mx \quad \text{for all } x \in [0, 1].$$

For $x > 0$ divide by $\sqrt{x}$ to get $1 \leq M\sqrt{x}$. Let $x \to 0^+$; the right-hand side tends to 0, contradiction. Hence no finite M works and $\sqrt{x}$ is not Lipschitz on $[0, 1]$.

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(b)

Assume without loss of generality that $x_1 < x_2$ (the case $x_2 < x_1$ is symmetric, and if $x_1 = x_2$, the result holds trivially with $\xi = x_1$).

Let $I = [x_1, x_2]$. Since $f \in C[a, b]$, the restriction of f to I is continuous, so $f(I)$ is a closed bounded interval $[m, M]$, where $m = \min_{x \in I} f(x)$ and $M = \max_{x \in I} f(x)$.

Let $K = \frac{c_1 f(x_1) + c_2 f(x_2)}{c_1 + c_2}$. Since $c_1 > 0$ and $c_2 > 0$, K can be written as $\alpha f(x_1) + (1 - \alpha) f(x_2)$, where $\alpha = \frac{c_1}{c_1 + c_2} \in (0, 1)$. Thus, K lies in the interval $[\min(f(x_1), f(x_2)), \max(f(x_1), f(x_2))]$, which is contained in $[m, M]$.

By the Intermediate Value Theorem (since $f(I)$ is connected and equals $[m, M]$), there exists $\xi \in I$ such that $f(\xi) = K$. Hence, ξ is between x_1 and x_2.

(c)

Consider $f(x) = x$ on $[0, 1]$, which is continuous. Let $x_1 = 0$, $x_2 = 1$, $c_1 = 1 > 0$, and $c_2 = -2 < 0$ (note that $c_1 \neq -c_2$).

Then,

$$\frac{c_1 f(x_1) + c_2 f(x_2)}{c_1 + c_2} = \frac{1 \cdot 0 + (-2) \cdot 1}{1 - 2} = \frac{-2}{-1} = 2.$$

On $[0, 1]$, $f(x) = x \in [0, 1]$, so there is no $\xi \in [0, 1]$ with $f(\xi) = 2$.